

Skills

Software: Python, C++, C, Java, Bash, MATLAB, XML/URDF, OOP, RTOS

Tools: PyTorch, scikit-learn, Docker, ROS2, Foxglove, Git, Linux, OpenCV, Redis, AWS S3, Jupyter Notebook, systemd

Concepts: Stereo Vision, Structure-from-Motion, Hardware Sensor Integration, PID Control, Frame Transforms

Technical Experience

Autonomy Core Member | *Watonomous Student Design Team*

Nov 2025 - Present | *Waterloo, ON, CA*

- Implementing sub-millisecond timing and synchronization accuracy between the computer, Velodyne LiDARs, and global shutter CMOS cameras by implementing PTP over Ethernet using a Novatel GPS as the grandmaster and STM32 as a hardware trigger

Systems Integration Engineering Intern | *Zoox (Amazon's Robotaxi)*

May - Aug 2024 | *Foster City, CA, USA*

- Integrated a system with CAN, 100Base-T1 automotive ethernet, and GMSL2 protocols to communicate with drive module ECUs, LiDAR, Radar, and camera sensors to enable the streamlined firmware flashing of 30+ control units at manufacturing
- Designed 2-6 layer PCBs with STM32, I2C, UART, JTAG, and USB(2.0) for low speed, 100Base-T1, GMSL2 communication using LT spice with schematic capture and PCB layout in Altium for power conversion, fault injection, logging, and breakout purposes

Robotics Software Engineering Intern | *Medra*

Jan - Apr & Sep - Dec 2023 | *San Francisco, CA, USA*

- Designed automation systems with a 7-DOF robot arm for bio-labs as the startup's 9th hire (\$52 million raised for Series A, 2025)
- Developed an initial simulation stack with collision detection using URDFs in Pybullet and further integrated sim tests in CircleCI
- Designed calibration and motion control tasks for labware interaction with a robotic arm to ~1mm repeatable accuracy and implemented real-time servo control with time-optimal path parameterization and trajectory smoothing using Ruckig
- Architected a Linux camera streaming software stack with latency compensation using Redis to increase perception reliability
- Identified labware using computer vision with OpenCV in Python to track workstation state with dynamic fault recovery

Projects

iPhone-based Open Source Autonomous Mobile Robot

- Turning an iPhone into a mobile robot that can run SLAM using ARKit, WASM, and Swift with custom hardware for a 2W differential robot using BLDC motors with FOC control and BLE communication

Estimating Depth from Monocular Vision using AI

- Investigated implementing and tuning various machine learning architectures (regression, support vector, MLP, CNN) using scikit-learn and PyTorch to evaluate the most accurate model for estimating mean depth of a scene from monocular vision

Full-stack Navigation of a Mobile Robot in Simulation

- Implemented mapping, planning, and control of a 2-w mobile robot using ROS2, Gazebo, and Foxglove, containerized in Docker
- Developed local costmap, global occupancy map, A* planning, and pure pursuit control ROS2 nodes from scratch in C++

Education

University of Waterloo, BAsC. Honours Mechatronics Engineering

Expected Apr 2026 | cGPA 92% / 3.96

Courses: Autonomous Mobile Robots, Foundations of AI, Multi-Sensor Data Fusion, Cooperative & Adaptive Algorithms

Undergraduate Research Assistant | *Vision and Image Processing (VIP) Research Lab*

Jan 2026 - Present

Researching visual-based SLAM navigation; supervised by Dr. John Zelek

Volunteer Research Assistant | *Emotionally Intelligent and Trustworthy Agents (EITA) Lab*

Sep 2025 - Present

Supporting research and literature review on intelligent social robotic tutoring systems for children; supervised by Dr. Moojan Ghafurian

Other Experience

Hardware Engineering Intern | *Verkada*

May - Aug 2025 | *San Mateo, CA, USA*

- Proposed a novel power-over-ethernet-based alerting device with full electrical and hardware architecture for the JDM model

Electrical Engineering Intern | *WHOOP*

Jan - May 2025 | *Boston, MA, USA*

- Focused on NFC wireless charging, power, and EE system optimization for WHOOP 5.0, WHOOP MG, and the Basic Charger